Audio Deep Learning

Assignment #1

Emotion Recognition Using Audio Features

Assignment overview and motivation

- *Format: Individual or group (max 3 students) with offline defense*
- *Grading: 17 points*
- *Timeline: Assignment # 1 available on 3rd June, Code submission deadline till 14th June, and offline defense deadline till 28th June*
- ***Submission via Google Form**: https://forms.gle/2nBFjH56BCrcDzX79*

While completing this assignment, you will develop and strengthen the following skills:

- *Approaching a real-world machine learning problem with custom naturalistic speech data (UA-SER dataset) and a target metric (Unweighted Average Recall - UAR).*
- *Conducting effective Exploratory Data Analysis (EDA) on audio signals and acoustic features (e.g., F0, MFCCs, energy) to understand paralinguistic cues.*
- *Designing a reliable, speaker-independent validation strategy to prevent data leakage and monitoring it throughout the entire development process.*
- *Using Self-Supervised Learning (SSL) speech models (such as wav2vec 2.0 and emotion2vec+) for solving Speech Emotion Recognition (SER) classification tasks.*
- *Performing a deep ablation study on different fine-tuning strategies (task transfer vs. language transfer), loss functions, and augmentation methods to pick the best approach.*

Task

Task Selection:

- *Use the Ukrainian Emotion Recognition dataset*

EDA and Metric Analysis:

- *Perform Exploratory Data Analysis (EDA).*

- *Analyze the proposed target metric (UAR): discuss its strengths, weaknesses, and potential edge cases. TIP: Check metrics from the next subsection: "Compare with the Dataset Author's work"*
- *Suggest complementary metrics to capture other aspects of performance.*
- *Propose an alternative target metric, with justification.*

Validation:

- *Use native train/test split. Design a separate validation split from the train set*
- *Explain your motivation for the chosen strategy and the importance of balancing gender and emotion classes within the splits.*
- *Discuss the challenges of achieving balanced class distributions in naturalistic emotion datasets.*

Spectral Features (MelSpec, MFCC):

- *Extract handcrafted acoustic features (e.g., MFCCs, Pitch/F0, RMS energy, Zero-Crossing Rate).*
- *Explore classical machine learning classifiers or simple neural networks to establish a baseline for the 4-class emotion classification problem.*
- *Collect all relevant results:*
    - *metrics (UAR, Accuracy, Macro F1)*
    - *hard samples analysis (e.g., structural variations between neutral and sad, or angry and happy).*
- *Conclude whether acoustic features alone are adequate.*
- *Prepare an ablation study on the tested models*

Self-Supervised Approach (Data2Vec, Wave2Vec, ...):

- *Explore pre-trained self-supervised speech approaches, contrasting language-transfer models (e.g., wav2vec2-xls-r-300m-uk) with task-transfer models (e.g., emotion2vec+).*
- *Collect all relevant results:*
    - *metrics (focusing on Unweighted Average Recall)*
    - *hard samples analysis.*
- *Conclude whether the model is adequate.*
- *Prepare an ablation study on the tested models, exploring freezing depths (e.g., number of frozen transformer blocks), loss functions (Focal Loss vs. Cross-Entropy), and data augmentation techniques.*

Compare with the Dataset Author's work

- *Explore the approach proposed by Olha Havryluk*
- *Explain how your approach differs from baseline and why is it better*

Final Report

- *Prepare a comprehensive report including:*
    - *data insights,*
    - *feature engineering,*
    - *tried algorithms and models,*
    - *chosen hyperparameters,*
    - *metrics,*
    - *insights and conclusions.*
- *Reports may be prepared in LaTeX or Markdown (README.md).*

NOTES

- *Do not hesitate to include failed ideas or experiments in your report. Documenting unsuccessful attempts is a strong indication that you systematically explored different approaches and gradually arrived at the best-performing solution, rather than relying on a "lucky shot." Moreover, analyzing failed approaches can often lead to valuable insights about the dataset, the task itself, and the behavior of deep learning algorithms. Such reflections may also highlight potential directions for future improvements and demonstrate a deeper understanding of the problem-solving process.*
- *Keep your EDA concise. Avoid overloading it with excessive plots, histograms, or tables. Focus only on key aspects of the data (e.g., missing values, target distribution). If a part of your EDA does not lead to meaningful conclusions, you likely don't need it (for example, full pairwise correlation matrices). Do NOT follow the typical Kaggle-style EDA — most of them are far too verbose and unfocused.*
- *Prepare your presentation carefully. A messy speech with too much code or too many files makes it nearly impossible for the audience to follow. Think in advance about what you want to highlight*

Grading Policy and Defense

Grading Policy is outlined in the table

Criterion	***Points***
Overall correctness and completeness of the submitted code for the lab assignment, overall quality (metrics) of the proposed solution	*5*
Practical question about the submitted code during defense	*7*
Theoretical part during defense (2 questions + 1 exercise)	*5*
Criterion	***Points***

The defense will be conducted during a practical session, where the Lecturer will evaluate your code, report, and presentation skills. It is important to emphasize that the assessment will not be limited to metrics, analysis, and the performance of your models, but will also take into account the overall quality of your code, including readability, structure, and adherence to good practices. During the defense, the Lecturer will evaluate each student's individual contribution, demonstrated skills, and depth of knowledge. In addition, up to 2 theoretical questions may be asked; these will be based on the questions outlined in Appendix 1, but will not necessarily be restricted to them.

Exercise with practical tasks similar to Appendix 2 will be evaluated in the form of a written test on one of the practical sessions

AI Tools usage policy

Scope

This policy applies to all generative AI tools that produce or modify text, code, images, or other digital content. This includes, but is not limited to:

- *Text generation tools: ChatGPT, Claude, Gemini, Mistral, LLaMA-based models, etc.*
- *Code generation tools: GitHub Copilot, OpenAI Codex, TabNine, etc.*
- *Language assistance tools: Grammarly, DeepL, Google Translate, etc.*
- *Local LLMs or self-hosted AI models.*

Allowed Uses

You may use AI tools for the following purposes, provided they comply with the conditions below:

- *Spell Check & Style Correction. Example: Using Grammarly to correct grammar and improve style in a lab report or paper review.*
- *Translation. Example: Using DeepL or Google Translate to translate sections of a document into another language.*
- *Idea Generation & Pitching. Example: Brainstorming possible approaches for solving a problem or generating research ideas.*
- *Self-Education. Example: Using ChatGPT to explain an unfamiliar algorithm or concept.*
- *Library & Documentation Exploration. Example: Asking AI to summarize PyTorch documentation for a specific class or function.*
- *Code Assistance (Partial). Writing or correcting specific functions or methods (not entire solutions). Examples: Write a function that reads data and converts it to another format. Implement a data collator with padding to the longest sequence. Create an image augmentation function for training.*

Conditions for Use

Full Transparency:

- *Any use of AI tools must be disclosed in your submission.*
- *Include all prompts, responses, and/or screenshots showing your interaction with the AI tool.*
- *If using online tools, include chat links (if available) or full transcripts in an appendix of your report.*

Partial Contribution Only:

- *AI tools should support your work — they must not produce the complete assignment, lab, or report.*
- *You must be able to explain and defend any AI-generated content during oral defense or Q&A*

Examples

Acceptable Uses:

- *“Correct grammar and style in this paragraph of my report.”*
- *“Write a function to load images from a directory and resize them.”*
- *“Suggest 3 ideas for improving model generalization.”*

Unacceptable Uses:

- *“Write the entire peer review for this paper.”*
- *“Generate the complete solution code for this assignment task.”*

Appendix 1 (Theoretical Questions)

- *Explain the difference between categorical and dimensional representations of emotion. What are the pros and cons of each in the context of SER?*
- *What is the circumplex model of affect, and how do valence and arousal map to specific discrete emotions (e.g., anger, happiness, sadness, neutral)?*
- *Explain the concept of cross-modal validation in dataset construction. How can lexical content (LLMs) or dimensional acoustic models (VAD) be used to validate human categorical emotion annotations?*
- *What are the primary sources of inter-annotator disagreement in naturalistic emotional speech? Why is moderate agreement (e.g., Fleiss' kappa ~0.40-0.60) considered acceptable in this field?*
- *What are the main acoustic feature families used in SER (prosodic, spectral, voice quality)? Give examples of features in each family and explain what type of emotional information they capture.*
- *Why is it necessary to apply loudness normalization (e.g., EBU R128 / LUFS) before feeding audio into an emotion recognition model?*
- *What types of audio data augmentations (e.g., pitch shift, time stretch, noise injection) are appropriate for SER without destroying the underlying emotional signal? Give examples where augmentation might harm the model's performance.*
- *Why is Unweighted Average Recall (UAR) preferred over standard accuracy in SER tasks, especially for naturalistic datasets?*
- *What is speaker identity leakage in the context of SER, and why must train/test splits be strictly speaker-independent?*
- *Why might focal loss or class-weighted cross-entropy be more effective than standard cross-entropy when fine-tuning on an imbalanced or highly confusable naturalistic emotion dataset?*
- *Explain the core idea behind self-supervised learning (SSL) in speech processing.*
- *What is the difference between contrastive masked prediction (e.g., wav2vec 2.0) and predictive self-distillation (e.g., data2vec/emotion2vec+) objectives?*
- *Explain the difference between "task transfer" and "language transfer" when fine-tuning SSL models for low-resource SER. Based on the UA-SER experiments, which approach yielded better results and why?*
- *Explain the role of an attention pooling layer compared to mean pooling when aggregating frame-level hidden states for utterance-level emotion classification.*
- *What is transfer learning, and what are its potential failure modes?*
- *When should we prefer full fine-tuning over feature extraction with a frozen backbone?*

Appendix 2 (Practical Exercises)

1.

Assume a discrete power spectrum $P(k)$ for frequency bins $k \in \{1, 2, 3, 4, 5\}$:

$$P = [10, 20, 30, 20, 10]$$

Triangular filter weights:

- $H_1(k)$: $[0.5, 1.0, 0.5, 0, 0]$
- $H_2(k)$: $[0, 0, 0.5, 1.0, 0.5]$

1. Compute the filterbank energy $E_1 = \sum_{k=1}^{5} P(k)H_1(k)$.
2. Compute the filterbank energy $E_2 = \sum_{k=1}^{5} P(k)H_2(k)$.
3. If the final Mel-Spectrogram feature is computed as $S_m = \log_{10}(E_m)$, calculate S_1 and S_2.

2.

Given hidden states h_1, h_2 and attention weight vector w:

$$h_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad h_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The attention weights α_t and final embedding z are computed as:

$$\alpha_t = \frac{\exp(w^T h_t)}{\sum_{t'=1}^{2} \exp(w^T h_{t'})}, \quad z = \sum_{t=1}^{2} \alpha_t h_t$$

1. Compute the unnormalized attention scores $s_1 = w^T h_1$ and $s_2 = w^T h_2$.
2. Compute the normalized attention weights α_1 and α_2.
3. Write the expression for the final utterance embedding $z \in \mathbb{R}^2$.
4. Conceptually, which frame (h_1 or h_2) will dominate the final utterance embedding and why?

3.

Given the l_2-normalized vectors:

$$c_t = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad q_t = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, \quad q_{neg} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The contrastive loss is defined as:

$$L_m = -\log \frac{\exp(\text{sim}(c_t, q_t)/\kappa)}{\exp(\text{sim}(c_t, q_t)/\kappa) + \exp(\text{sim}(c_t, q_{neg})/\kappa)}$$

where $\text{sim}(u, v) = u^T v$ and $\kappa = 0.1$.

1. Compute the cosine similarity $\text{sim}(c_t, q_t)$.
2. Compute the cosine similarity $\text{sim}(c_t, q_{neg})$.
3. Compute the final contrastive loss L_m.